Database System Engineering and Distributed Backend Development
Course Code: 25CS1302E

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SQL VIEWS

1. CREATE DATABASE

```
CREATE DATABASE BankDB2;
USE BankDB2;
```

2	20:47:25	USE BankDB2

2. CREATE CUSTOMER TABLE

```sql
CREATE TABLE Customer(
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15),
    Email VARCHAR(100),
    City VARCHAR(50)
);


CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
```

Output

Action Output

#	Time	Action
2	20:47:25	USE BankDB2
3	20:47:34	CREATE TABLE Customer (Customer_ID INT PRIMARY

3. CREATE ACCOUNT TABLE

```sql
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2),
    Branch VARCHAR(50),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

Output

Action Output

#	Time	Action
2	20:47:25	USE BankDB2
3	20:47:34	CREATE TABLE Customer (Customer_ID INT PRIMAF
4	20:47:43	CREATE TABLE Customer (Customer_ID INT PRIMAF
5	20:47:54	CREATE TABLE Account (Account_No INT PRIMAR

4. CREATE BANK TRANSACTION TABLE

```sql
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (Account_No)
    REFERENCES Account(Account_No)
);
```

Output

Action Output

#	Time	Action	Message
9	20:48:54	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VA...	Error Code: 1050. Table 'customer' already exis
10	20:49:03	CREATE TABLE Customer(Customer_ID INT PRIMARY KEY, Customer_Name VA...	Error Code: 1050. Table 'customer' already exis
11	20:49:16	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, ...	Error Code: 1050. Table 'account' already exist
12	20:51:37	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INC...	0 row(s) affected

5. CREATE LOAN TABLE

```sql
CREATE TABLE Loan (
    Loan_ID INT PRIMARY KEY,
    Customer_ID INT,
    Loan_Type VARCHAR(30),
    Loan_Amount DECIMAL(12,2),
    Interest_Rate DECIMAL(5,2),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

Output

Action Output

#	Time	Action	Message
10	20:49:03	CREATE TABLE Customer(Customer_ID INT PRIMARY KEY, Customer_Name VA...	Error Code: 1050. Ta
11	20:49:16	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, ...	Error Code: 1050. Ta
12	20:51:37	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INC...	0 row(s) affected
13	20:52:06	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_...	0 row(s) affected

6. INSERT CUSTOMER DATA

```sql
INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, Email, City)
VALUES
(101, 'vaishnavi', '9876543210',
 'ravi@gmail.com', 'Hyderabad'),

(102, 'likhitha', '9876543211',
 'priya@gmail.com', 'Vijayawada'),

(103, 'Arjun Reddy', '9876543212',
 'arjun@gmail.com', 'Bangalore'),

(104, 'Sneha Rao', '9876543213',
```

Output

Action Output

#	Time	Action	Message
11	20:49:16	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, ...	Error Code: 1050. Table 'acc
12	20:51:37	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INC...	0 row(s) affected
13	20:52:06	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_...	0 row(s) affected
14	20:52:58	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (1...	10 row(s) affected Records:

7. INSERT ACCOUNT DATA

```sql
INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),

(10002, 102, 'Savings', 75000, 'Vijayawada'),

(10003, 103, 'Current', 120000, 'Bangalore'),

(10004, 104, 'Savings', 45000, 'Chennai'),

(10005, 105, 'Current', 90000, 'Hyderabad'),

(10006, 106, 'Savings', 65000, 'Delhi'),

(10007, 107, 'Current', 150000, 'Mumbai'),
```

Output

Action Output

#	Time	Action	Message
12	20:51:37	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INC...	0 row(s) affected
13	20:52:06	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_...	0 row(s) affected
14	20:52:58	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (1...	10 row(s) affected Records: 10 D
15	20:53:32	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) V...	10 row(s) affected Records: 10 D

8. INSERT TRANSACTION DATA

```sql
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),

(10001, 'WITHDRAW', 5000),

(10002, 'DEPOSIT', 15000),

(10003, 'WITHDRAW', 20000),

(10004, 'DEPOSIT', 5000),

(10005, 'WITHDRAW', 10000),
```

Output

Action Output

#	Time	Action	Message
13	20:52:06	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_...	0 row(s) affected
14	20:52:58	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (1...	10 row(s) affected Records: 10 Dupli
15	20:53:32	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) V...	10 row(s) affected Records: 10 Dupli
16	20:54:02	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (1...	11 row(s) affected Records: 11 Dupli

9. INSERT LOAN DATA

```sql
INSERT INTO Loan
(Loan_ID, Customer_ID, Loan_Type,
 Loan_Amount, Interest_Rate)
VALUES
(501, 101, 'Home Loan', 5000000, 7.5),

(502, 102, 'Education Loan', 1000000, 6.5),

(503, 103, 'Car Loan', 800000, 8.2),
```

Output

Action Output

#	Time	Action	Message
14	20:52:58	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (1...	10 row(s) affe
15	20:53:32	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) V...	10 row(s) affe
16	20:54:02	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (1...	11 row(s) affe
17	20:54:39	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)...	8 row(s) affec

10. DISPLAY ORIGINAL TABLES

```
SELECT * FROM Customer;

SELECT * FROM Account;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Customer_ID	Customer_Name	Phone	Email	City
101	vaishnavi	9876543210	ravi@gmail.com	Hyderabad
102	likhitha	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
106	Anil Kumar	9876543215	anil@gmail.com	Delhi
107	Meena Reddy	9876543216	meena@gmail.com	Mumbai
108	Rahul Sharma	9876543217	rahul@gmail.com	Pune
109	Lakshmi Devi	9876543218	lakshmi@gmail.com	Hyderabad
110	Suresh Babu	9876543219	suresh@gmail.com	Vijayawada

Customer 1

```
SELECT * FROM Account;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	65000.00	Delhi
10007	107	Current	150000.00	Mumbai
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad
10010	110	Current	110000.00	Vijayawada

Account 2

```
SELECT * FROM Loan;
```

Result Grid | Filter Rows: | Edit: | Export/

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
505	105	Home Loan	4000000.00	7.20
506	106	Car Loan	900000.00	8.50
507	107	Business Loan	3000000.00	9.00
508	109	Personal Loan	600000.00	10.00
NULL	NULL	NULL	NULL	NULL

11. CREATE A VIEW FOR ALL CUSTOMERS

```
CREATE VIEW Customer_View AS
SELECT *
FROM Customer;

SELECT * FROM Customer_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	Phone	Email	City
101	vaishnavi	9876543210	ravi@gmail.com	Hyderabad
102	likhitha	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
106	Anil Kumar	9876543215	anil@gmail.com	Delhi
107	Meena Reddy	9876543216	meena@gmail.com	Mumbai
108	Rahul Sharma	9876543217	rahul@gmail.com	Pune

HIGH BALANCE ACCOUNT VIEW

```sql
CREATE VIEW High_Balance_View AS
SELECT
    Account_No,
    Customer_ID,
    Account_Type,
    Balance
FROM Account
WHERE Balance > 100000;
SELECT * FROM High_Balance_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Account_No	Customer_ID	Account_Type	Balance
10003	103	Current	120000.00
10007	107	Current	150000.00
10010	110	Current	110000.00

High_Balance_View 5

CUSTOMER + ACCOUNT VIEW

```sql
CREATE VIEW Customer_Account_View AS
SELECT
    C.Customer_ID,
    C.Customer_Name,
    C.City,
    A.Account_No,
    A.Account_Type,
    A.Balance,
    A.Branch
FROM Customer C
JOIN Account A
ON C.Customer_ID = A.Customer_ID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch
101	vaishnavi	Hyderabad	10001	Savings	50000.00	Hyderabad
102	likhitha	Vijayawada	10002	Savings	75000.00	Vijayawada
103	Arjun Reddy	Bangalore	10003	Current	120000.00	Bangalore
104	Sneha Rao	Chennai	10004	Savings	45000.00	Chennai
105	Kiran Kumar	Hyderabad	10005	Current	90000.00	Hyderabad
106	Anil Kumar	Delhi	10006	Savings	65000.00	Delhi
107	Meena Reddy	Mumbai	10007	Current	150000.00	Mumbai

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TOTAL BANK BALANCE

```sql
CREATE VIEW Total_Bank_Balance AS
SELECT
    SUM(Balance) AS Total_Balance
FROM Account;

SELECT * FROM Total_Bank_Balance;
```

Result Grid | Filter Rows: | Export:

Total_Balance
825000.00

TRANSACTION DETAILS WITH CUSTOMER

```sql
SELECT * FROM Total_Bank_Balance;
CREATE VIEW Customer_Transaction_View AS
SELECT
    C.Customer_Name,
    A.Account_No,
    A.Account_Type,
    T.Transaction_ID,
    T.Transaction_Type,
    T.Amount,
    T.Transaction_Date
FROM Customer C
JOIN Account A
ON C.Customer_ID = A.Customer_ID
JOIN Bank_Transaction T
ON A.Account_No = T.Account_No;
SELECT * FROM Customer_Transaction_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_Name	Account_No	Account_Type	Transaction_ID	Transaction_Type	Amount	Transaction_Date
vaishnavi	10001	Savings	2	WITHDRAW	5000.00	2026-08-16 20:54:02
likhitha	10002	Savings	3	DEPOSIT	15000.00	2026-08-16 20:54:02
Arjun Reddy	10003	Current	4	WITHDRAW	20000.00	2026-08-16 20:54:02
Sneha Rao	10004	Savings	5	DEPOSIT	5000.00	2026-08-16 20:54:02
Kiran Kumar	10005	Current	6	WITHDRAW	10000.00	2026-08-16 20:54:02
Anil Kumar	10006	Savings	7	DEPOSIT	12000.00	2026-08-16 20:54:02
Meena Reddy	10007	Current	8	DEPOSIT	25000.00	2026-08-16 20:54:02
Rahul Sharma	10008	Savings	9	WITHDRAW	5000.00	2026-08-16 20:54:02

Customer_Transaction_View 8

Create a view that displays:

Customer_ID

Customer_Name
Account_No
Account_Type
Balance

for all customers.

```sql
CREATE VIEW CustomerAccountView AS
SELECT
    c.Customer_ID,
    c.Customer_Name,
    a.Account_No,
    a.Account_Type,
    a.Balance
FROM Customer c
JOIN Account a
    ON c.Customer_ID = a.Customer_ID;

SELECT * FROM CustomerAccountView;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	Account_No	Account_Type	Balance
101	vaishnavi	10001	Savings	50000.00
102	likhitha	10002	Savings	75000.00
103	Arjun Reddy	10003	Current	120000.00
104	Sneha Rao	10004	Savings	45000.00
105	Kiran Kumar	10005	Current	90000.00
106	Anil Kumar	10006	Savings	65000.00
107	Meena Reddy	10007	Current	150000.00
108	Rahul Sharma	10008	Savings	35000.00

Create a view that displays all Current Accounts from Hyderabad.

```sql
CREATE VIEW HyderabadCurrentAccounts AS
SELECT *
FROM Account
WHERE Account_Type = 'Current'
  AND Branch = 'Hyderabad';


  SELECT * FROM HyderabadCurrentAccounts;
```

ult Grid | Filter Rows: | Export: | Wrap Cell Content:

Account_No	Customer_ID	Account_Type	Balance	Branch
10005	105	Current	90000.00	Hyderabad